



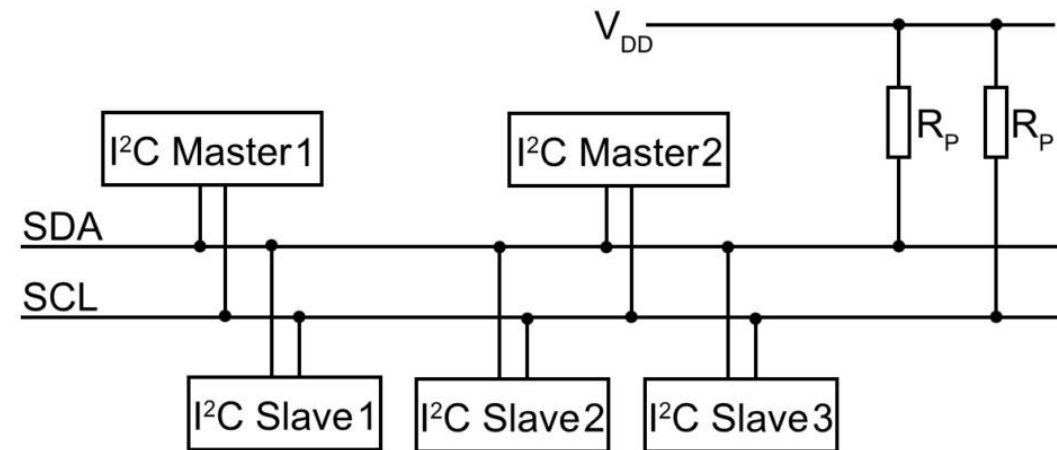
# Synchronous Serial Communication

Communication Protocols

# Synchronous Serial Communication (I<sup>2</sup>C)

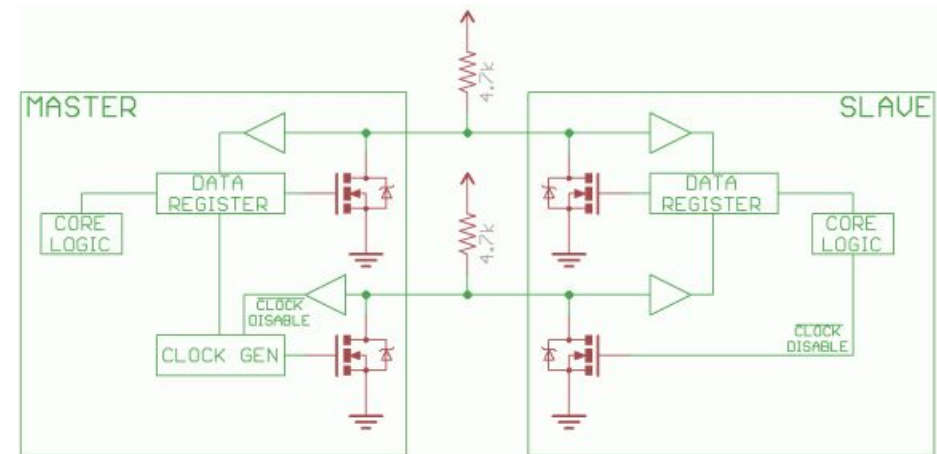
## ❖ Inter-Integrated Circuit (I<sup>2</sup>C)

- A synchronous serial communication protocol
- Allows multiple slaves to communicate with one or more masters
- Is used in short-distance communications
- Uses two signal lines
  - **SCL**: Serial Clock which is always generated by the current master
  - **SDA**: Serial Data
- Supports half-duplex communication
- Speed: upto 3.4 mbps, typically 400 kbps
- Drivers of the lines are **open drain**
  - They can only pull the signal line low
  - Pull-up resistors are required (1-10 k $\Omega$ )



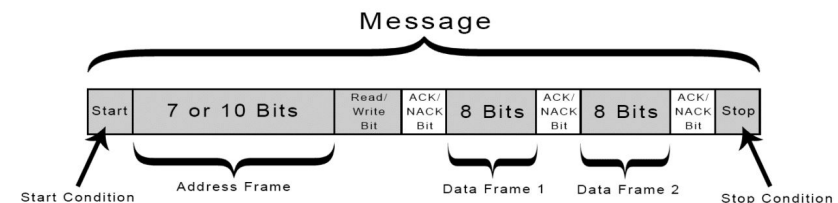
# Synchronous Serial Communication (I<sup>2</sup>C - Signal Levels)

- ❖ I2C supply voltage, typically ranging from 1.2 V to 5.5 V
- ❖ Flexibility in connecting devices with different voltage levels
  - Because the devices on the bus don't drive the signals high
  - The pull-up resistors are connected to the lower voltage
  - E.g. a 5V master and a 3.3V slave (pull-up resistors should be connected to 3.3V)
  - If the voltage difference is too great, a level shifter is required
- ❖ For short distances, use 4.7 k $\Omega$  pull-up resistors and for long distances, or systems with lots of devices, use smaller resistors
- ❖ Note that slaves can disable the master clock



# Synchronous Serial Communication (I<sup>2</sup>C - Message)

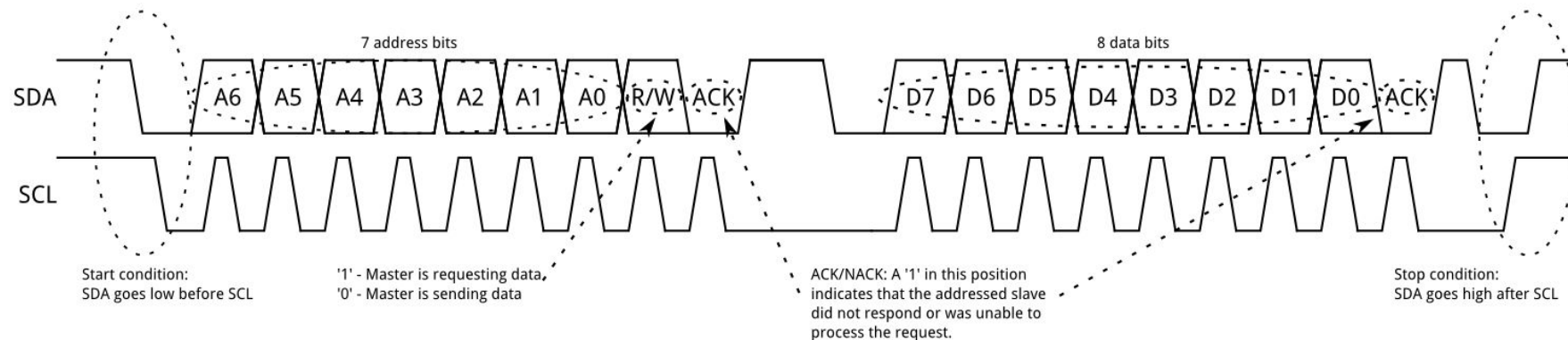
- ❖ A message has two types of frame:
  - An **address frame** which is the **unique address** of the slave
    - I<sup>2</sup>C supports up to 112 slaves in the 7 bits and 1008 slaves in the 10 bits addressing
    - 16 addresses are reserved (1111 XXX and 0000 XXX)
  - One or more **data frames**: 8-bit data messages passed from master to slave or vice versa.
- ❖ Every message is started with the **Start Condition**, then
  - Followed by the **Address Frame**, then
  - The **Read/Write** bit (1: read and 0: write) indicating the operation, then
  - The **NACK/ACK** bit (1: NACK, 0: ACK), then
  - The data frames followed by the NACK/ACK bit and
  - The **Stop Condition**
- ❖ The ACK is sent by the receiver for all frames (address and data)



# Synchronous Serial Communication (I<sup>2</sup>C - Signaling)

## ❖ Start Condition

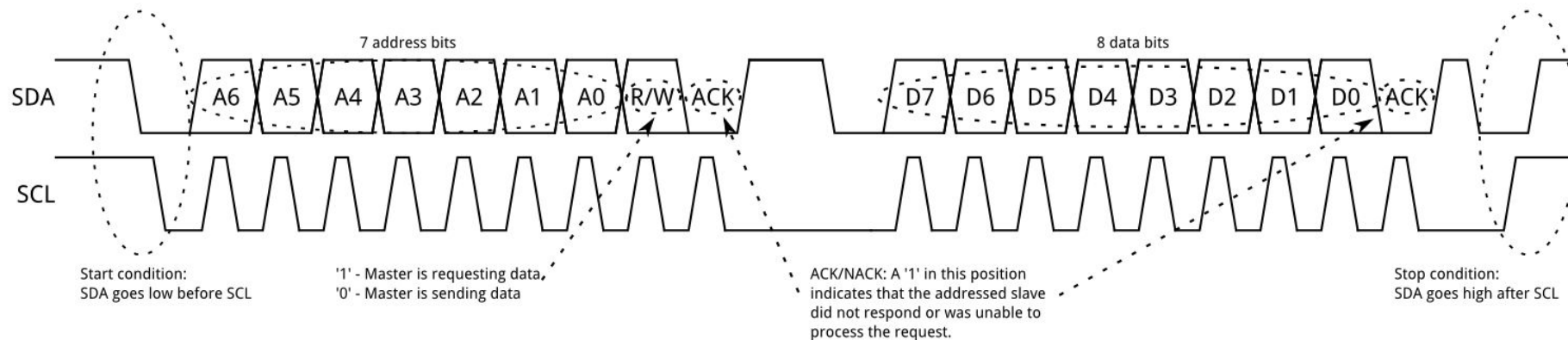
- The master leaves SCL high and pulls SDA low.
  - This puts all slave devices on notice that a transmission is about to start.
- If several masters want to take ownership of the bus at the same time
  - 0 is dominant and 1 is recessive.
  - Bit monitoring and an arbitration process is used to solve conflicts
- It is possible to have repeated starts



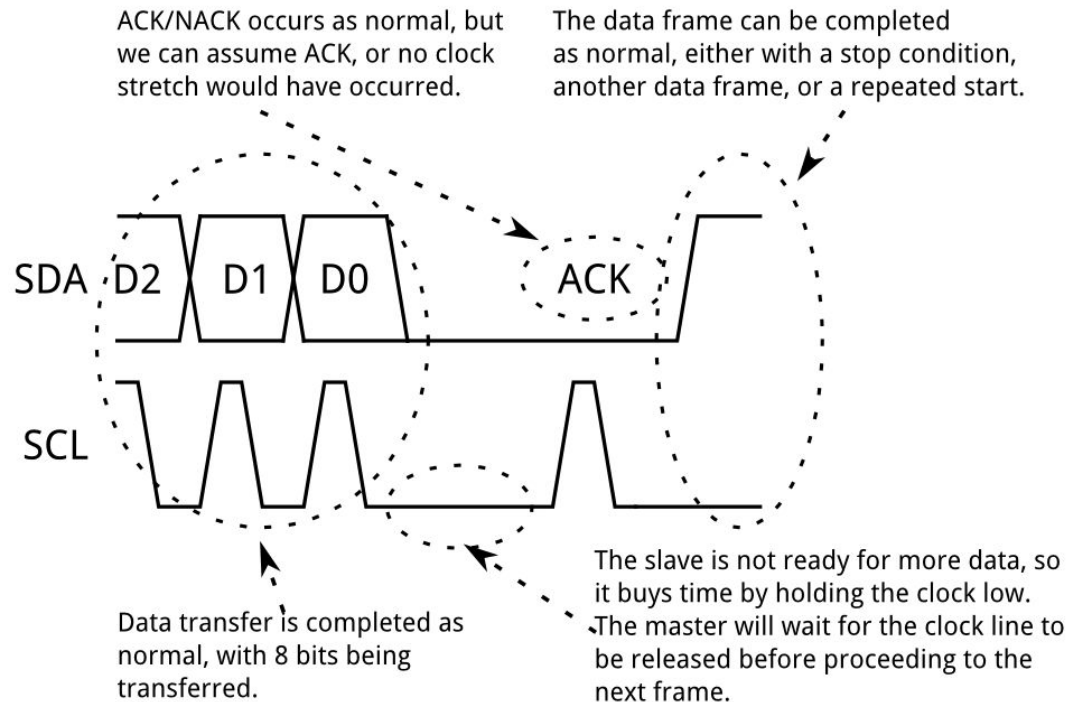


# Synchronous Serial Communication (I<sup>2</sup>C - Signaling)

- ❖ Data is placed on the SDA after SCL goes low, and is sampled after the SCL goes high
  - All changes on data line is done during clock low, except **Start** and **Stop** Conditions
- ❖ The most significant bit (MSB) of every frame (address and data) is sent first
- ❖ The receiver after receiving every frame, acknowledges the frame by pulling the SDA low
- ❖ Stop condition
  - Once all the data frames have been sent, the **master** generates a stop condition
  - Is a **low to high** transition on SDA after a **low to high** transition on SCL, with SCL remaining high.

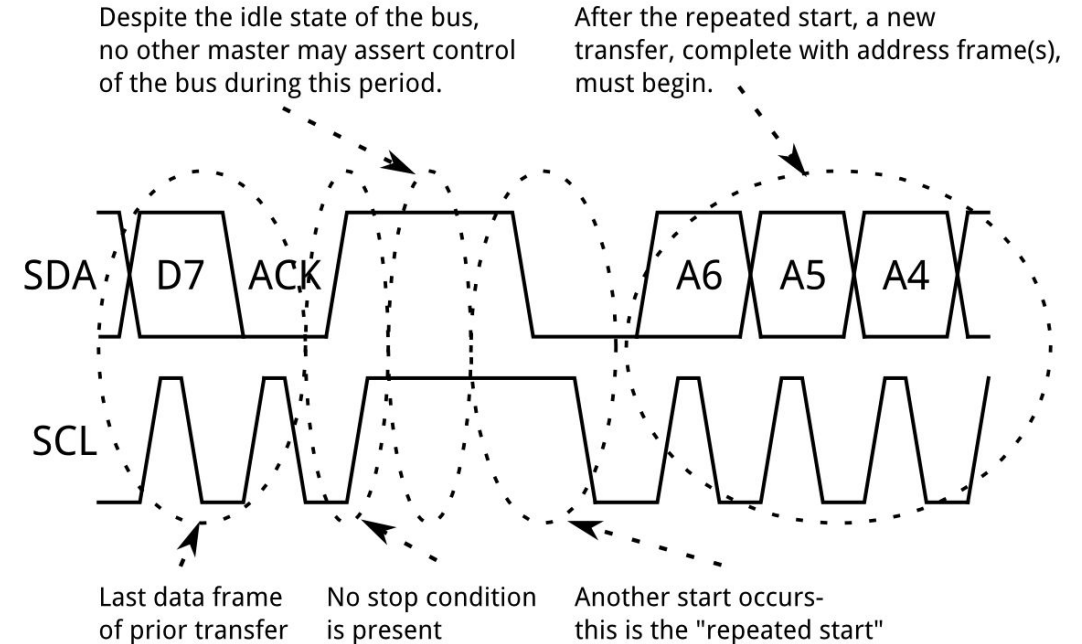


# Synchronous Serial Communication (I<sup>2</sup>C - Signaling)



## Clock Stretching

It is used when the master's data rate will exceed the slave's ability to process the data.

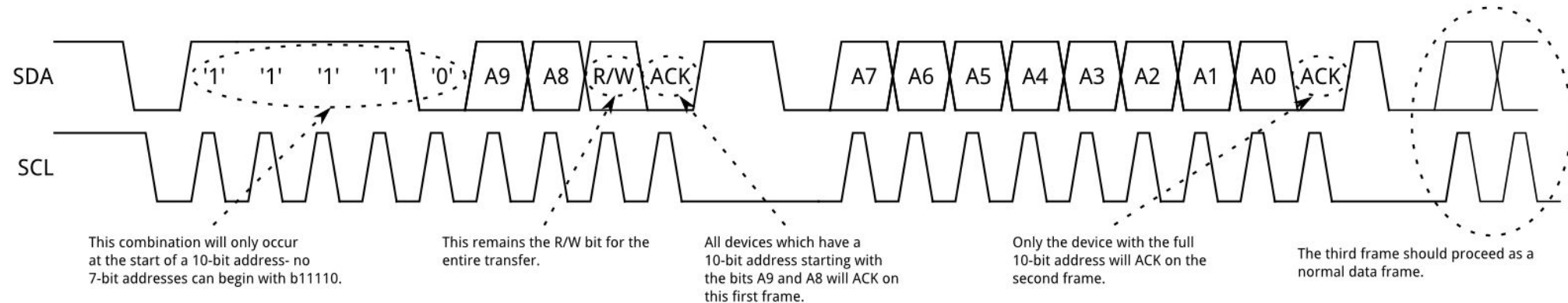


## Repeated Start Conditions

It is used when a master wants to exchange several messages without losing the control of the bus.

# Synchronous Serial Communication (I<sup>2</sup>C - Signaling)

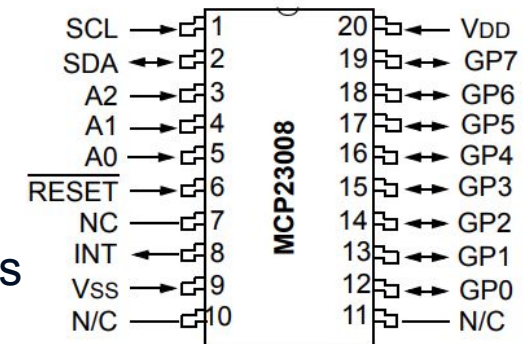
## ❖ 10-bit Addresses



## ❖ Note that 10-bit address devices can coexist with 7-bit address devices

- The leading '11110' part of the address is not a part of any valid 7-bit addresses

## ❖ An I<sup>2</sup>C IO expander like MCP23008 can be used to expand the GPIOs





# Synchronous Serial Communication (I<sup>2</sup>C)

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## ❖ Some useful links

- [I2C Bus, Interface and Protocol](#)
- [Sparkfun I2C Tutorial](#)
- [How I2C Communication Works?](#)
- [Basics of the I2C Communication Protocol](#)
- [What is I2C, Basics for Beginners](#)
- [I2C Protocol Tutorial | How I2C Protocol works](#)